

Motor control - Solutions

1. Even after amputation, the **internal models** of the limb remain intact in the brain. The motor cortex, premotor areas, cerebellum, and parietal regions still contain representations of the missing limb and the learned mappings between motor commands and predicted sensory feedback.

When a person intends to move the phantom limb, the brain issues a normal motor command and generates an *effference copy* using the forward model. The forward model predicts the expected sensory feedback (proprioception, touch, etc.), even though no actual feedback arrives from the missing limb. The absence of real sensory feedback does not immediately invalidate the prediction, so the motor system still “believes” the limb moved. This mismatch between predicted and received feedback contributes to the vivid perception that the limb is still present.

Thus, phantom limb sensations arise because internal models and motor commands remain functional in the absence of peripheral feedback. The brain’s predictive machinery continues to generate sensory experiences and movement intentions as if the limb were still there.

2. We can frame the decision using expected loss (or expected cost), where the rational choice minimizes expected loss.

Compute the true probability of an accident. From the statistics:

$$P(\text{accident}) = \frac{15}{3.3 \text{ million}} \approx 4.5 \times 10^{-6}.$$

Hence,

$$P(\text{safe}) = 1 - P(\text{accident}) \approx 0.9999955.$$

Define the loss function.

$$L(\text{accident}) = 100,000, \quad L(\text{safe}) = -1.$$

(We use negative for pleasure, so lower values are better.)

Compute expected loss for jumping.

$$E[L_{\text{jump}}] = P(\text{safe}) \cdot L(\text{safe}) + P(\text{accident}) \cdot L(\text{accident}).$$

$$E[L_{\text{jump}}] = 0.9999955 \cdot (-1) + 4.5 \times 10^{-6} \cdot (100,000) = -0.9999955 + 0.45 = -0.55.$$

Compare to not jumping. If you *don't jump*, you avoid risk but lose the opportunity:

$$E[L_{\text{no jump}}] = 0 \quad (\text{neutral, no pleasure, no risk}).$$

Decision. Because $E[L_{\text{jump}}] < E[L_{\text{no jump}}]$, the expected loss is smaller for jumping (it's even slightly negative, meaning a net expected gain).

Normatively, you *should jump*: the pleasure outweighs the risk once you correct for your overestimated probability.

However, your *subjective prior* over risk is biased: if you still believe $P(\text{accident}) = 0.1$ instead of the true 4.5×10^{-6} , with that belief:

$$E[L_{\text{jump}}] = 0.9 \cdot (-1) + 0.1 \cdot (100,000) = -0.9 + 10,000 = 9,999.1,$$

so you definitely wouldn't jump.

Conclusion. The example illustrates how prior beliefs about probabilities influence decision-making. When subjective priors overweight risk (as in anxiety), expected loss increases dramatically, leading to overly cautious or avoidance behavior — even when objective evidence suggests the risk is minimal. This reflects the same Bayesian principle: decisions depend on how prior beliefs are combined with evidence.

3. We define a loss function $f(e)$ that describes how costly an error of size e is. For each movement, the total loss is the sum of the losses across the two attempts:

$$L_{\text{total}} = f(e_1) + f(e_2).$$

You can think of this as your brain's "cost" for inaccuracy.

We have two options:

$$(a) \ e_1 = 2, \ e_2 = 2 \quad (b) \ e_1 = 4, \ e_2 = 0.$$

Depending on the *shape* of the loss function, you will prefer one pattern over the other.

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If choice (a) is better – large errors are punished strongly (convex loss). Let the loss be quadratic:

$$f(e) = e^2.$$

Then:

$$L_a = f(2) + f(2) = 2^2 + 2^2 = 4 + 4 = 8, \quad L_b = f(4) + f(0) = 4^2 + 0^2 = 16 + 0 = 16.$$

If choice (b) is better – success on one attempt matters most (concave or step loss).

If the loss function values “at least one hit,” we can model it as a step-like or concave function.

Example 1: Step-like (hit/miss) or binary loss

$$f(e) = \begin{cases} 0 & \text{if hit } (e = 0) \\ 1 & \text{if miss } (e > 0) \end{cases}$$

Then:

$$L_a = 1 + 1 = 2, \quad L_b = 1 + 0 = 1.$$

Here, $L_b < L_a$, so (b) is better — you prefer one perfect hit to two small misses.

Example 2: Concave loss (square root of the error)

$$f(e) = \sqrt{e}.$$

Then:

$$L_a = \sqrt{2} + \sqrt{2} = 2\sqrt{2} \approx 2.83, \quad L_b = \sqrt{4} + \sqrt{0} = 2 + 0 = 2.$$

If both choices are equally good – linear loss. If the loss grows linearly with error magnitude:

$$f(e) = e,$$

then:

$$L_a = 2 + 2 = 4, \quad L_b = 4 + 0 = 4.$$

4. We model the (unknown) target position as a random variable x .

$$\text{Prior: } x \sim \mathcal{N}(\mu_p, \sigma_p^2), \quad \text{Observation: } z \sim \mathcal{N}(x, \sigma_\ell^2).$$

Here, $\mu_p = 0$, $\sigma_p^2 = 4$, and on this trial $z = 3$ with sensory noise $\sigma_\ell^2 = 1$.

Bayes' rule:

$$P(x | z) \propto P(z | x) P(x).$$

We use “ $\propto$ ” because the denominator $P(z)$ is a normalization constant that does not depend on x .

Write both Gaussians (ignore their front constants, which also do not depend on x):

$$P(x) \propto \exp\left(-\frac{(x - \mu_p)^2}{2\sigma_p^2}\right), \quad P(z | x) \propto \exp\left(-\frac{(z - x)^2}{2\sigma_\ell^2}\right).$$

Multiply the two Gaussians:

$$P(x | z) \propto \exp\left(-\frac{(x - \mu_p)^2}{2\sigma_p^2} - \frac{(z - x)^2}{2\sigma_\ell^2}\right).$$

Expand the squared terms:

$$(x - \mu_p)^2 = x^2 - 2x\mu_p + \mu_p^2, \quad (z - x)^2 = x^2 - 2xz + z^2.$$

$$P(x | z) \propto \exp\left(-\frac{1}{2}\left[x^2\left(\frac{1}{\sigma_p^2} + \frac{1}{\sigma_\ell^2}\right) - 2x\left(\frac{\mu_p}{\sigma_p^2} + \frac{z}{\sigma_\ell^2}\right) + \text{constants}\right]\right).$$

Match to the Gaussian form, which has the general form:

$$\exp\left[-\frac{1}{2\sigma^2}(x^2 - 2x\mu + \mu^2)\right].$$

By comparing coefficients, we identify:

$$\frac{1}{\sigma_{\text{post}}^2} = \frac{1}{\sigma_p^2} + \frac{1}{\sigma_\ell^2}, \quad \frac{\mu_{\text{post}}}{\sigma_{\text{post}}^2} = \frac{\mu_p}{\sigma_p^2} + \frac{z}{\sigma_\ell^2}.$$

Hence,

$$\sigma_{\text{post}}^2 = \frac{1}{\frac{1}{\sigma_p^2} + \frac{1}{\sigma_\ell^2}}, \quad \mu_{\text{post}} = \frac{\frac{\mu_p}{\sigma_p^2} + \frac{z}{\sigma_\ell^2}}{\frac{1}{\sigma_p^2} + \frac{1}{\sigma_\ell^2}}.$$

For reliable vision ($\sigma_\ell^2 = 1$):

$$\mu_{\text{post}} = \frac{0.25 \cdot 0 + 1 \cdot 3}{0.25 + 1} = 2.4 \text{ cm}, \quad \sigma_{\text{post}}^2 = \frac{1}{1.25} = 0.8 \text{ cm}^2.$$

For noisy vision ($\sigma_\ell^2 = 9$):

$$\mu_{\text{post}} = \frac{0.25 \cdot 0 + 0.111 \cdot 3}{0.25 + 0.111} \approx 0.92 \text{ cm}, \quad \sigma_{\text{post}}^2 = \frac{1}{0.361} \approx 2.77 \text{ cm}^2.$$

4.3. When vision is more reliable (small σ_ℓ^2) than the prior, the posterior mean is pulled toward the visual observation. When vision is noisy (large σ_ℓ^2), the posterior mean shifts toward the prior. This reflects Bayesian integration: people weigh prior knowledge more when sensory evidence is uncertain.